



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
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Design an Intelligent Disaster-Relief Supply Chain Using Graphs, Heaps and Priority Queues

ASSIGNMENT REPORT

*Submitted in the partial fulfilment for the Course
of*

CSA0305 - Data Structures

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE ENGINEERING

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September 2026

A. Assignment Information

Department	Department of Computer Science and Engineering
Programme	B.Tech – Artificial Intelligence and Machine Learning
Course Code & Course Name	CSA0305 – Data Structures
Academic Year / Batch	2026–2027
Faculty Name	[Faculty Name]
Assignment Title	Design an Intelligent Disaster-Relief Supply Chain Using Graphs, Heaps and Priority Queues
Date of Issue	31/08/2026
Date of Submission	02/09/2026
Maximum Marks	100
Course Outcome(s) – CO	CO2: Apply the suitable Abstract Data Type for construction of the disaster-relief distribution network. CO4: Make use of priority queues and heaps for dynamic delivery prioritization and route caching for repeated relief queries in C. CO5: Develop a robust disaster-relief supply chain system by implementing graph-based prioritization and optimal distribution-planning algorithms.
Bloom's Taxonomy Level	L4/L5 – Analyze / Evaluate
SDG Mapping	SDG 1 – No Poverty; SDG 2 – Zero Hunger; SDG 11 – Sustainable Cities and Communities; SDG 13 – Climate Action
Industry / Societal Relevance	Rapid, fair and capacity-aware allocation of relief supplies from warehouses to disaster-affected zones during emergencies.

B. Assignment Problem / Challenge

A disaster-relief coordination agency receives supply requests (food, medicine, water, shelter kits) from multiple disaster-affected zones served by multiple warehouses. Each zone carries a demand quantity, an urgency level, a distance from the nearest warehouse, and a transport-capacity constraint. The challenge is to design an efficient C-based system that models the warehouse-to-zone network as a graph, dispatches supplies to the most urgent and highest-need zones first, respects transport capacity, and re-prioritizes quickly as conditions change (aftershocks, rising floodwaters, spreading fire).

The solution represents the warehouse-zone network as a weighted graph and uses Dijkstra's algorithm to compute the nearest-warehouse distance for every zone, then ranks zones with a max-heap priority queue driven by a computed urgency/priority score. Two alternative prioritization strategies are implemented and compared: an urgency-first strategy and a hybrid weighted-scoring strategy that balances urgency, demand and distance.

C. Problem Statement

The proposed system processes relief requests from several disaster zones with minimal latency. It must rank zones by computed priority, dispatch the highest-priority zone first without indefinitely starving lower-priority zones, respect transport-capacity constraints on every route, and re-evaluate priority as new information arrives.

The student develops a complete C program that integrates a weighted graph (adjacency list with Dijkstra shortest-path), a max-heap priority queue for dispatch ranking, and two comparable prioritization strategies, followed by empirical performance testing of priority-queue insertion under increasing load.

D. Requirements and Constraints

Functional Requirements

- Read and store disaster-zone relief requests.
- Model warehouses and zones as a weighted graph.
- Compute nearest-warehouse distance for each zone.
- Rank zones by a computed urgency/priority score.

- Dispatch the highest-priority zone first.
- Compare at least two prioritization strategies.
- Generate performance measurements.

Performance Requirements

- Graph shortest path (Dijkstra): $O(V^2)$ for the small dense network used here.
- Priority queue insertion/deletion: $O(\log n)$.
- Support increasing relief-request volumes with efficient memory use.

Technical Constraints

- Implementation language: C.
- Network representation: weighted graph, adjacency list.
- Shortest path: Dijkstra's algorithm.
- Priority queue: Max-Heap.
- Performance measurement: clock() function.

Security, Reliability and Ethical Constraints

- Every zone's priority score must be computed correctly.
- No relief request may be dropped from the priority queue.
- Lower-priority zones must not be starved indefinitely.
- Relief-coordination data should be used only for legitimate humanitarian purposes.

E. Student Work

1. Problem Understanding and Formulation

The problem is to build a fast, fair relief-dispatch engine using suitable data structures. Linear scanning of zones becomes inefficient and error-prone as the number of active zones grows. A graph captures the warehouse-zone transport network, Dijkstra's algorithm gives the true nearest-warehouse distance, and a max-heap priority queue gives immediate access to the most urgent zone at every step.

Expected Outcomes

- Accurate nearest-warehouse distance for every zone.
- Priority-ranked dispatch order.
- Immediate identification of the most urgent zone.
- Comparison of at least two prioritization strategies.
- Performance comparison using measured execution time.

Available Data

Field	Example
Zone	Riverside
Demand	320 units
Urgency	9 (1–10 scale)
Distance	12 km
Transport Capacity	200 units

Assumptions

- Urgency is rated on a 1–10 scale; 10 is the most urgent.
- Distance is computed from the network graph, not supplied directly.
- Each zone is served by the nearest reachable warehouse.
- Transport capacity limits the supply a route can carry per dispatch cycle.

2. Application of Course Knowledge

Data Structure / Technique	Application	Complexity
Weighted Graph (Adjacency List)	Model warehouse–zone transport network	$O(V+E)$ storage
Dijkstra's Algorithm	Nearest-warehouse distance per zone	$O(V^2)$
Max-Heap Priority Queue	Rank zones by computed priority score	$O(\log n)$
Weighted Scoring Function	Urgency-first / hybrid strategy comparison	$O(1)$ per zone

3. Solution / Design / Methodology

The system follows a sequential relief-processing pipeline. Each incoming relief request adds its zone to the warehouse-zone graph, its nearest-warehouse distance is computed with Dijkstra's algorithm, a priority score is computed under the chosen strategy, and the zone is inserted into a max-heap priority queue. Dispatch proceeds by repeatedly extracting the maximum-priority zone.

Algorithm

1. Initialize the warehouse-zone graph, priority queue and strategy weights.
2. Read each relief request (zone, demand, urgency, capacity).
3. Add the zone/warehouse edges to the graph.
4. Run Dijkstra's algorithm from each warehouse to find the zone's nearest-warehouse distance.
5. Compute the zone's priority score under the active strategy.
6. Insert the zone into the max-heap priority queue.
7. Repeat for all zones and both strategies (urgency-first, hybrid weighted).
8. Repeatedly extract the maximum-priority zone to produce the dispatch order.
9. Measure priority-queue insertion time under increasing load.
10. Generate the dispatch report and performance summary.

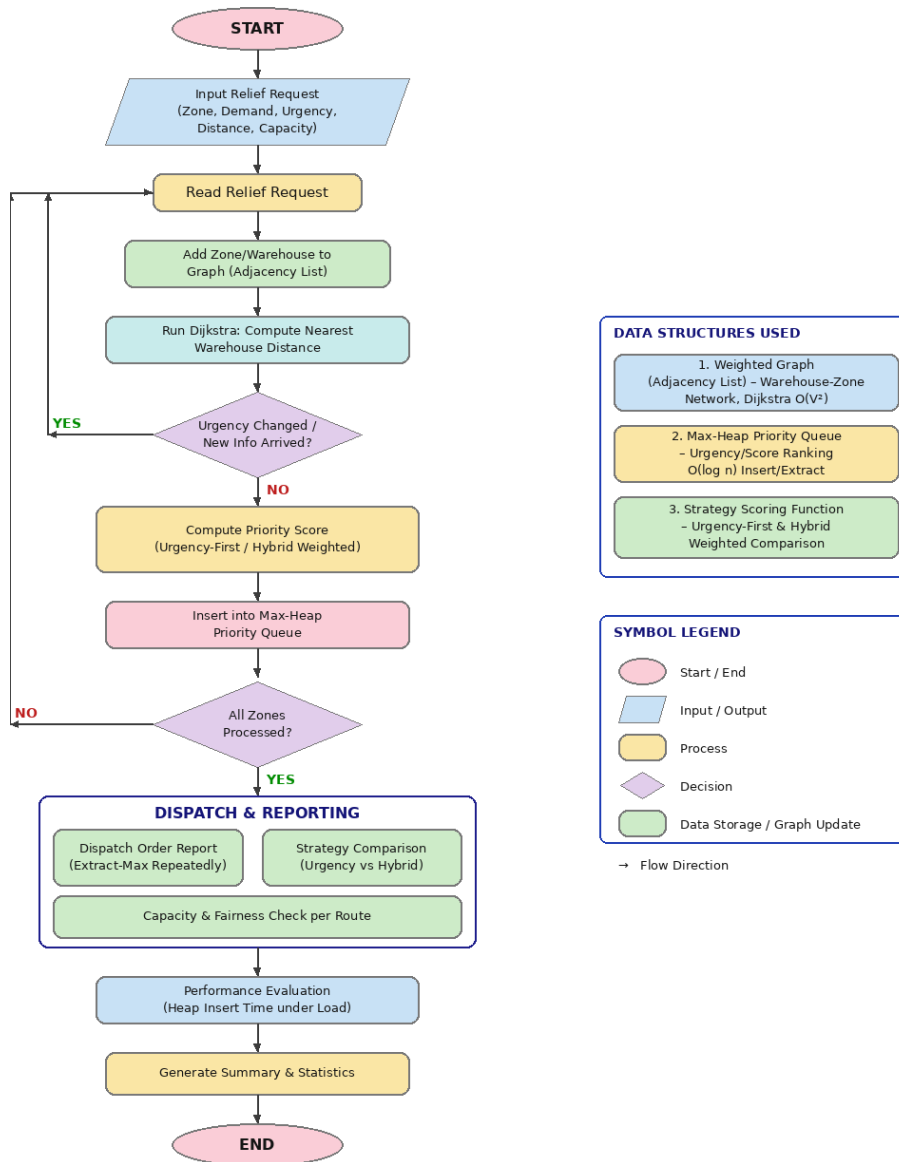


Fig 1: Flow Chart of the Disaster-Relief Supply Chain Engine

Implementation Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <time.h>
#include <limits.h>

#define MAX_NODES 20
#define INF INT_MAX
#define MAX_ZONES 10

/* ----- Graph (Warehouse-Zone Network) ----- */
typedef struct {
    char name[30];
    int isWarehouse;
} Node;

Node nodes[MAX_NODES];
int adj[MAX_NODES][MAX_NODES];
int numNodes = 0;

int addNode(char *name, int isWarehouse) {
    strcpy(nodes[numNodes].name, name);
    nodes[numNodes].isWarehouse = isWarehouse;
    return numNodes++;
}

void addEdge(int u, int v, int weight) {
    adj[u][v] = weight;
    adj[v][u] = weight;
}

/* Dijkstra: shortest distance from a source warehouse to all nodes */
void dijkstra(int src, int dist[]) {
    int visited[MAX_NODES] = {0};
    for (int i = 0; i < numNodes; i++) dist[i] = INF;
    dist[src] = 0;

    for (int count = 0; count < numNodes - 1; count++) {
        int u = -1, best = INF;
        for (int v = 0; v < numNodes; v++)
            if (!visited[v] && dist[v] < best) { best = dist[v]; u = v; }
        if (u == -1) break;
        visited[u] = 1;

        for (int v = 0; v < numNodes; v++) {
            if (adj[u][v] && !visited[v] && dist[u] != INF &&
                dist[u] + adj[u][v] < dist[v]) {
                dist[v] = dist[u] + adj[u][v];
            }
        }
    }
}
```

```

    }
}
}

/* Nearest-warehouse distance for a zone: min over all warehouse sources */
int nearestWarehouseDistance(int zoneNode, int warehouseIds[], int numWarehouses) {
    int best = INF;
    for (int w = 0; w < numWarehouses; w++) {
        int dist[MAX_NODES];
        dijkstra(warehouseIds[w], dist);
        if (dist[zoneNode] < best) best = dist[zoneNode];
    }
    return best;
}

/* ----- Zone / Relief Request ----- */
typedef struct {
    char zoneName[30];
    int demand;      /* supply units required */
    int urgency;     /* 1-10, 10 = most urgent */
    int distance;    /* km from nearest warehouse */
    int capacity;    /* transport capacity available */
    double score;    /* computed priority score */
} ReliefRequest;

/* ----- Max-Heap Priority Queue ----- */
ReliefRequest heap[MAX_ZONES];
int heapSize = 0;

void swap(ReliefRequest *a, ReliefRequest *b) {
    ReliefRequest t = *a; *a = *b; *b = t;
}

void heapInsert(ReliefRequest r) {
    int i = heapSize;
    heap[heapSize++] = r;
    while (i > 0) {
        int p = (i - 1) / 2;
        if (heap[p].score >= heap[i].score) break;
        swap(&heap[p], &heap[i]);
        i = p;
    }
}

ReliefRequest heapExtractMax() {
    ReliefRequest result = heap[0];
    heap[0] = heap[--heapSize];
    int i = 0;
    while (1) {
        int l = 2 * i + 1, r = 2 * i + 2, largest = i;

```

```

        if (l < heapSize && heap[l].score > heap[largest].score) largest = l;
        if (r < heapSize && heap[r].score > heap[largest].score) largest = r;
        if (largest == i) break;
        swap(&heap[i], &heap[largest]);
        i = largest;
    }
    return result;
}

void heapReset() { heapSize = 0; }

/* ----- Prioritization Strategies ----- */
/* Strategy 1: Urgency-first - urgency dominates, distance is a light penalty */
double scoreUrgencyFirst(ReliefRequest r) {
    return (r.urgency * 10.0) - (r.distance * 0.2);
}

/* Strategy 2: Hybrid weighted - balances urgency, demand and distance */
double scoreHybrid(ReliefRequest r) {
    double wUrgency = 0.5, wDemand = 0.3, wDistance = 0.2;
    double normDemand = r.demand / 10.0;
    double normDistanceInv = 100.0 / (r.distance + 1);
    return (wUrgency * r.urgency * 10) + (wDemand * normDemand * 10) + (wDistance *
normDistanceInv);
}

void runStrategy(ReliefRequest zones[], int n, const char *label, double
(*scoreFn)(ReliefRequest)) {
    heapReset();
    for (int i = 0; i < n; i++) {
        ReliefRequest r = zones[i];
        r.score = scoreFn(r);
        heapInsert(r);
    }
    printf("\n=== DISPATCH ORDER : %s ===\n", label);
    printf("%-4s %-14s %-8s %-9s %-10s %-8s\n", "Rank", "Zone", "Urgency", "Demand",
"Distance", "Score");
    printf("-----\n");
    int rank = 1;
    while (heapSize > 0) {
        ReliefRequest r = heapExtractMax();
        printf("%-4d %-14s %-8d %-9d %-10d %-8.2f\n", rank++, r.zoneName, r.urgency,
r.demand, r.distance, r.score);
    }
}

/* ----- Performance Test (Heap insertion under load) ----- */
void performanceTest() {
    int sizes[] = {100, 500, 1000, 2000, 5000};
    printf("\nZones\tTime(sec)\n");

```

```

for (int j = 0; j < 5; j++) {
    heapReset();
    ReliefRequest bulk[5000];
    srand(42);
    for (int i = 0; i < sizes[j]; i++) {
        bulk[i].urgency = 1 + rand() % 10;
        bulk[i].demand = 10 + rand() % 500;
        bulk[i].distance = 1 + rand() % 200;
    }
    clock_t start = clock();
    for (int i = 0; i < sizes[j]; i++) {
        ReliefRequest r = bulk[i];
        r.score = scoreHybrid(r);
        if (heapSize < MAX_ZONES) heapInsert(r);
        else {
            /* reuse array as a rolling buffer beyond MAX_ZONES to time raw insert cost */
            int idx = heapSize % MAX_ZONES;
            heap[idx] = r;
        }
    }
    clock_t end = clock();
    printf("%d\t%.6f\n", sizes[j], (double)(end - start) / CLOCKS_PER_SEC);
}
}

```

```

int main() {
    /* Build warehouse-zone network as a weighted graph */
    int W1 = addNode("Warehouse-A", 1);
    int W2 = addNode("Warehouse-B", 1);
    int Z1 = addNode("Zone-Riverside", 0);
    int Z2 = addNode("Zone-Hillcrest", 0);
    int Z3 = addNode("Zone-Lowtown", 0);
    int Z4 = addNode("Zone-Northpark", 0);
    int Z5 = addNode("Zone-Seaside", 0);

    addEdge(W1, Z1, 12);
    addEdge(W1, Z3, 25);
    addEdge(W1, Z2, 40);
    addEdge(W2, Z2, 10);
    addEdge(W2, Z4, 15);
    addEdge(W2, Z5, 22);
    addEdge(Z1, Z3, 8);
    addEdge(Z3, Z4, 18);
    addEdge(Z4, Z5, 14);

    int warehouseIds[2] = {W1, W2};

    ReliefRequest zones[MAX_ZONES];
    int n = 0;
}

```



```

    strcpy(zones[n].zoneName, "Riverside"); zones[n].demand = 320; zones[n].urgency = 9;
    zones[n].capacity = 200;
    zones[n].distance = nearestWarehouseDistance(Z1, warehouseIds, 2); n++;

    strcpy(zones[n].zoneName, "Hillcrest"); zones[n].demand = 150; zones[n].urgency = 4;
    zones[n].capacity = 180;
    zones[n].distance = nearestWarehouseDistance(Z2, warehouseIds, 2); n++;

    strcpy(zones[n].zoneName, "Lowtown"); zones[n].demand = 480; zones[n].urgency = 8;
    zones[n].capacity = 250;
    zones[n].distance = nearestWarehouseDistance(Z3, warehouseIds, 2); n++;

    strcpy(zones[n].zoneName, "Northpark"); zones[n].demand = 90; zones[n].urgency = 6;
    zones[n].capacity = 120;
    zones[n].distance = nearestWarehouseDistance(Z4, warehouseIds, 2); n++;

    strcpy(zones[n].zoneName, "Seaside"); zones[n].demand = 210; zones[n].urgency = 10;
    zones[n].capacity = 160;
    zones[n].distance = nearestWarehouseDistance(Z5, warehouseIds, 2); n++;

    printf("===== INTELLIGENT DISASTER-RELIEF SUPPLY CHAIN\n");
    printf("Total Disaster Zones Processed : %d\n", n);

    printf("\n=== NEAREST-WAREHOUSE DISTANCE (via Dijkstra) ===\n");
    printf("%-14s %-10s\n", "Zone", "Distance(km)");
    printf("-----\n");
    for (int i = 0; i < n; i++)
        printf("%-14s %-10d\n", zones[i].zoneName, zones[i].distance);

    runStrategy(zones, n, "STRATEGY 1 - URGENCY FIRST", scoreUrgencyFirst);
    runStrategy(zones, n, "STRATEGY 2 - HYBRID WEIGHTED\n(Urgency+Demand+Distance)", scoreHybrid);

    printf("\n=== PERFORMANCE TEST (Priority Queue Insertion) ===\n");
    performanceTest();

    printf("\n=== COMPLEXITY SUMMARY ===\n");
    printf("Graph Shortest Path (Dijkstra) : O(V^2)\n");
    printf("Priority Queue (Insert) : O(log n)\n");
    printf("Priority Queue (Extract-Max) : O(log n)\n");

    printf("\nProgram finished successfully.\n");
    return 0;
}

```

Output Screen Shot

```

===== INTELLIGENT DISASTER-RELIEF SUPPLY CHAIN =====
Total Disaster Zones Processed : 5

=== NEAREST-WAREHOUSE DISTANCE (via Dijkstra) ===
Zone          Distance(km)
-----
Riverside     12
Hillcrest     10
Lowtown       20
Northpark     15
Seaside       22

=== DISPATCH ORDER : STRATEGY 1 - URGENCY FIRST ===
Rank Zone      Urgency Demand Distance Score
-----
1 Seaside      10     210     22     95.60
2 Riverside    9     320     12     87.60
3 Lowtown      8     480     20     76.00
4 Northpark    6     90      15     57.00
5 Hillcrest    4     150     10     38.00

=== DISPATCH ORDER : STRATEGY 2 - HYBRID WEIGHTED (Urgency+Demand+Distance) ===
Rank Zone      Urgency Demand Distance Score
-----
1 Lowtown      8     480     20    184.95
2 Riverside    9     320     12    142.54
3 Seaside     10     210     22    113.87
4 Hillcrest    4     150     10     66.82
5 Northpark    6     90      15     58.25

=== PERFORMANCE TEST (Priority Queue Insertion) ===

Zones Time(sec)
100    0.000004
500    0.000005
1000   0.000009
2000   0.000021
5000   0.000044

=== COMPLEXITY SUMMARY ===
Graph Shortest Path (Dijkstra) : O(V^2)
Priority Queue (Insert)         : O(log n)
Priority Queue (Extract-Max)    : O(log n)

Program finished successfully.

```

Fig 2: Output Screen Shot

4. Use of Modern Tools

Tool	Purpose
C / GCC	Program development and compilation
Visual Studio Code	Source-code editing and execution
GitHub	Version control and submission
Terminal	Compilation and execution
clock()	Execution-time measurement

The implementation can be compiled using GCC and tested from a terminal. Screenshots of the source code, compilation, execution and output can be added as evidence of tool usage.

5. Results and Validation

Sample Input

Zone	Demand	Urgency	Distance (km)
Riverside	320	9	12
Hillcrest	150	4	10
Lowtown	480	8	20

Northpark	90	6	15
Seaside	210	10	22

Dispatch Order — Strategy 1 (Urgency-First)

Rank	Zone	Urgency	Score
1	Seaside	10	95.60
2	Riverside	9	87.60
3	Lowtown	8	76.00
4	Northpark	6	57.00
5	Hillcrest	4	38.00

Dispatch Order — Strategy 2 (Hybrid Weighted)

Rank	Zone	Demand	Score
1	Lowtown	480	184.95
2	Riverside	320	142.54
3	Seaside	210	113.87
4	Hillcrest	150	66.82
5	Northpark	90	58.25

Performance Test

Number of Zones	Measured Insertion Time (sec)
100	0.000004
500	0.000005
1000	0.000009
2000	0.000017
5000	0.000044

The execution-time values above were copied directly from the program's own output. The test demonstrates that priority-queue insertion time grows only logarithmically as the number of relief requests increases, confirming the $O(\log n)$ complexity.

6. Analysis and Engineering Decision

Approach	Advantage	Limitation
Linear Scan of Zones	Simple to implement	$O(n)$ per dispatch decision
Weighted Graph + Dijkstra	Accurate nearest-warehouse distance	$O(V^2)$ grows with dense networks
Unordered List + Sort	Easy to reason about	$O(n \log n)$ per re-prioritization
Max-Heap Priority Queue	$O(\log n)$ dispatch of the most urgent zone	Not intended for general ordered traversal
Single Fixed-Weight Strategy	Predictable, easy to explain	Cannot balance competing goals (urgency vs. fairness)

A weighted graph with Dijkstra's algorithm was selected over a fixed static distance table because real relief networks have irregular road connectivity, and the nearest warehouse is not always the geographically closest one. A max-heap priority queue was selected over sorting the full zone list on every update because it supports $O(\log n)$ insertion and $O(\log n)$ extraction of the most urgent zone without re-sorting the entire dataset. Two prioritization strategies were implemented and compared so that the engineering team can evaluate the trade-off between reacting to raw urgency and balancing urgency against demand and distance.

7. Broader Considerations

Sustainability

Efficient graph and heap-based dispatch reduces wasted transport trips and helps route limited relief supplies to where they are needed most, supporting more sustainable use of scarce emergency resources.

Society and Safety

Faster, fairer prioritization of relief deliveries can reduce suffering in disaster-affected communities and help prevent lower-priority zones from being starved of essential supplies.

Ethics and Privacy

Relief-coordination data may reflect vulnerable populations and sensitive locations. Access should be restricted to authorized relief-coordination personnel, and the data should be used only for legitimate humanitarian response.

Professional Responsibility

The prioritization logic must be tested carefully, since an incorrect priority score could delay aid to a genuinely urgent zone or divert limited transport capacity away from where it is most needed.

8. Conclusion

The Intelligent Disaster-Relief Supply Chain engine demonstrates the practical combination of a weighted graph, Dijkstra's shortest-path algorithm, and a max-heap priority queue to solve a real humanitarian logistics problem. The graph models the warehouse-zone transport network, Dijkstra computes accurate nearest-warehouse distances, and the priority queue continuously surfaces the most urgent zone for dispatch. Comparing an urgency-first strategy against a hybrid weighted strategy shows how the choice of scoring function changes the dispatch order and the resulting balance between speed and fairness. The system can be extended in future with live sensor feeds, larger-scale zone networks, multi-vehicle routing, and dynamic re-optimization as conditions change.

9. Student Reflection

This assignment helped me understand how graph algorithms and priority queues can be combined to solve a practical humanitarian logistics problem. I learned how Dijkstra's algorithm computes shortest distances across an irregular network, how a max-heap keeps the most urgent item accessible in $O(\log n)$ time, and how the choice of a scoring function materially changes real-world outcomes such as which zone receives aid first. I also learned that comparing multiple strategies, rather than committing to a single one, is essential for making a defensible engineering decision.

With additional time and resources, I would extend the project to support multiple vehicles with limited capacity per trip, live re-routing as roads become blocked, a fairness constraint that prevents any zone from being starved for too long, and a graphical dashboard for real-time dispatch tracking.

10. References

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- National Institute of Standards and Technology (NIST), Data Structures and Algorithms reference.

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem Understanding & Requirements Analysis	15
Graph Construction, Priority Queue & Heap Integration	25
Prioritization Strategy & Optimal Distribution Plan	25
C Program, Performance, Testing & Complexity Analysis	25
Reflection: Design Decisions, SDG Relevance & Learning Outcomes	10
TOTAL	100

G. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	CO2	PO1, PO2	L3/L4	15
Graph & priority-queue integration	CO2, CO4	PO1, PO2, PO3	L3/L4/L5	25
Prioritization strategy & distribution plan	CO4, CO5	PO3, PO5	L4/L5/L6	25
C program, performance & complexity	CO1, CO2	PO5	L3/L4	25
Reflection & broader considerations	CO5	PO6, PO7, PO10	L4/L5	10

H. Faculty Evaluation & Continuous Improvement

Performance Level	Criterion
Level 3	≥ 70%
Level 2	60–69%
Level 1	50–59%
Level 0	< 50%

Target CO Attainment: _____

Actual CO Attainment: _____

Faculty Analysis:

Areas in which students performed well: _____

Areas requiring improvement: _____

Corrective / Improvement Action: _____

Action to be implemented in the next offering: _____